Master's Seminar: Artificial Intelligence - Visual Computing
3D Gaussian Splatting

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Why 3D Gaussian Splatting?

- **The NeRF Dilemma**
  NeRFs are either slow and provide photorealistic scenes or they are reasonably fast but have to sacrifice image quality. Real-time applications are not possible.

- **3D Gaussians as a Novel Approach**
  Using 3D Gaussians as the explicit base primitive, and abandoning neural networks was the key to unlocking previously unseen rendering speeds.

- **Impact on 3D Reconstruction**
  Real-time rendering of reconstructed scenes represents a fundamental disruption in 3D Reconstruction fascilitating many new downstream applications.

Overview

1 Background

2 3D Gaussian Data Structure

3 Training Procedure
- Point Cloud Initialization
- Forward Pass
- Backward Pass

4 Novel-View Synthesis

5 Summary and Discussion

Core Concepts

3D Scene Creation

Novel-View Synthesis

Splatting

Definitions

Creates 3D environments such as individual objects or outdoor spaces from an overlapping set of 2D input images.

Core Concepts

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Definitions

Synthesizes new 2D imagery from the 3D scene using new camera positions.

Real-time capability requires at least 24–30 frames per second (FPS) to appear smooth enough for the human eye.

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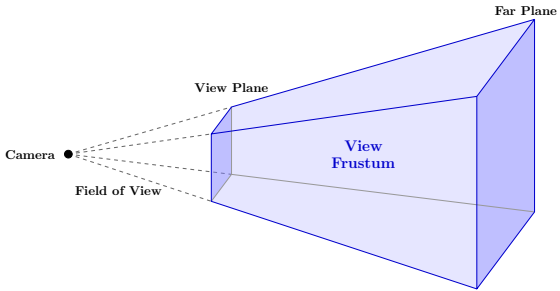
Definitions

The process of moving all elements inside the view frustum onto the view plane preserving order, adding to aggregated color and opacity of each pixel.



Core Innovations

Origin & Architecture

The Base Primitive

Fully Differentiable

Direct Editability

Hardware Optimization

Details

Introduced by Kerbl et al. at INRIA in France in late 2023, 3D Gaussian Splatting (3DGS) combines existing concepts to create a learnable explicit radiance field representation.

The publication had a very high impact in the field with over 10,000 citations in three years.

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A 3D Gaussian is an ellipsoid "fuzzy blob" that loses opacity towards its edges and can take shapes ranging from perfect spheres to elongated needles or flat shapes.

Using them as base primitives bypasses previous issues with explicit fields and allows for arbitrary positional precision.

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By cleverly decomposing the covariance matrix, 3D Gaussians become differentiable and eligible for standard gradient descent backpropagation.

Most importantly, this means 3DGS operates completely without neural networks.

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Unlike the neural network "black box" of NeRFs, 3D Gaussians provide direct editability of the scene.

This makes scene modifications, such as additions or lighting changes, highly accessible for downstream applications.

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Kerbl et al. mapped their architecture exhaustively onto modern GPU hardware.

The centerpiece is the tile-based rasterizer, which allows real-time novel-view synthesis at 1080p resolution while matching or improving upon state-of-the-art NeRF quality.

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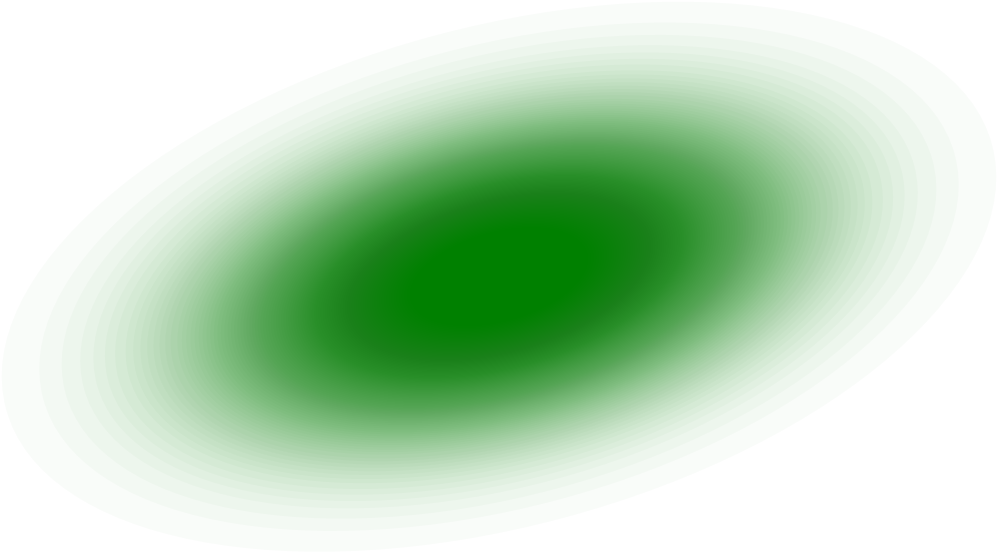

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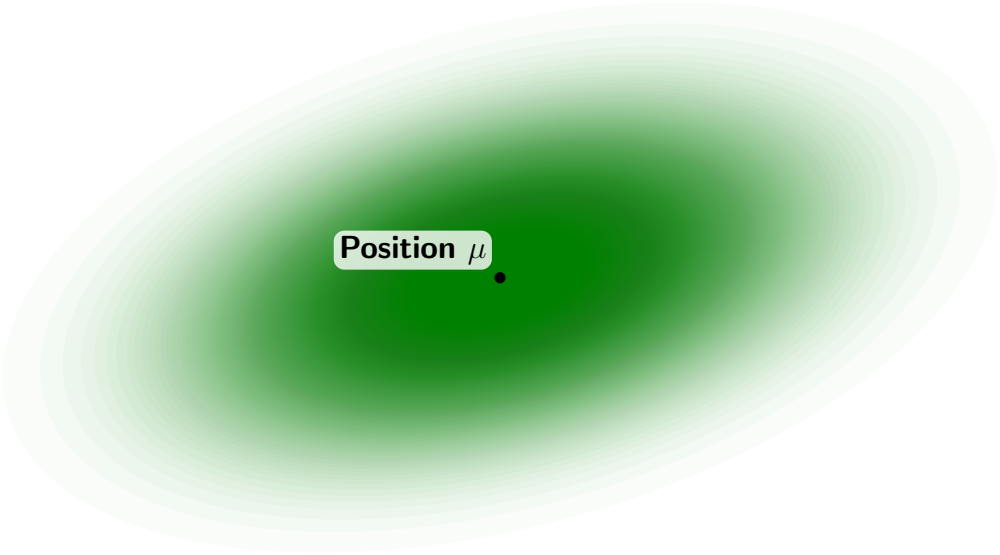

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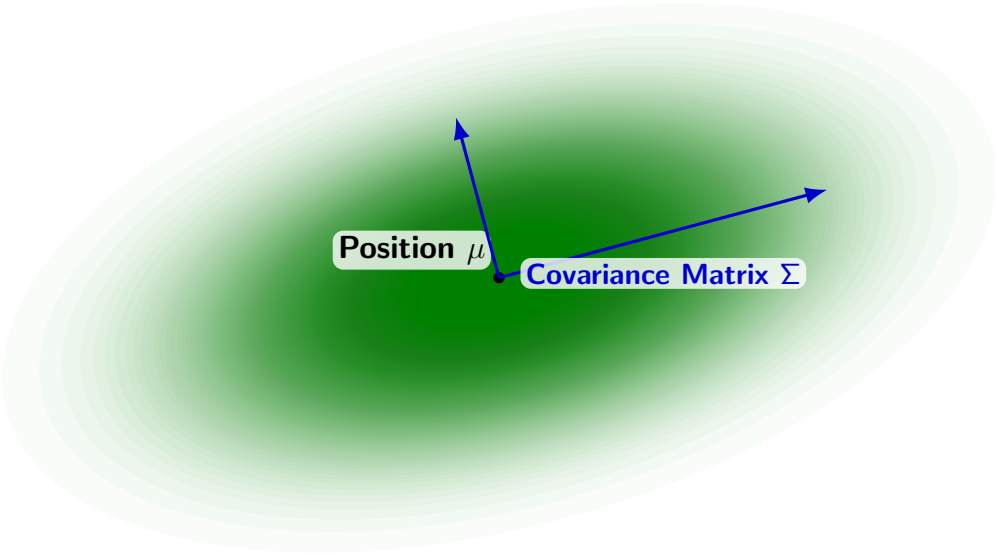

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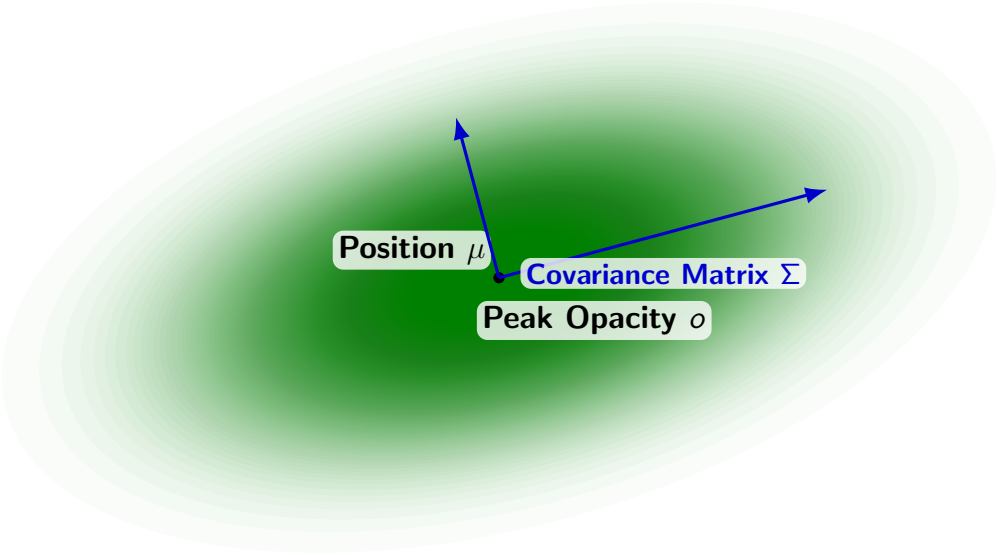

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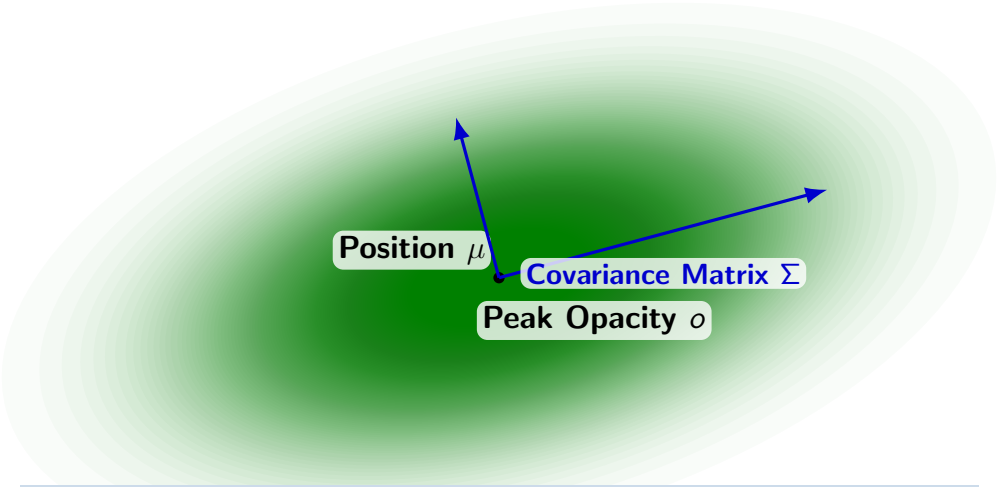

Multivariate Normal Distribution: $\alpha(p) = o \cdot exp\left(-\frac{1}{2}(p-\mu)^T\Sigma^{-1}(p-\mu)\right)$

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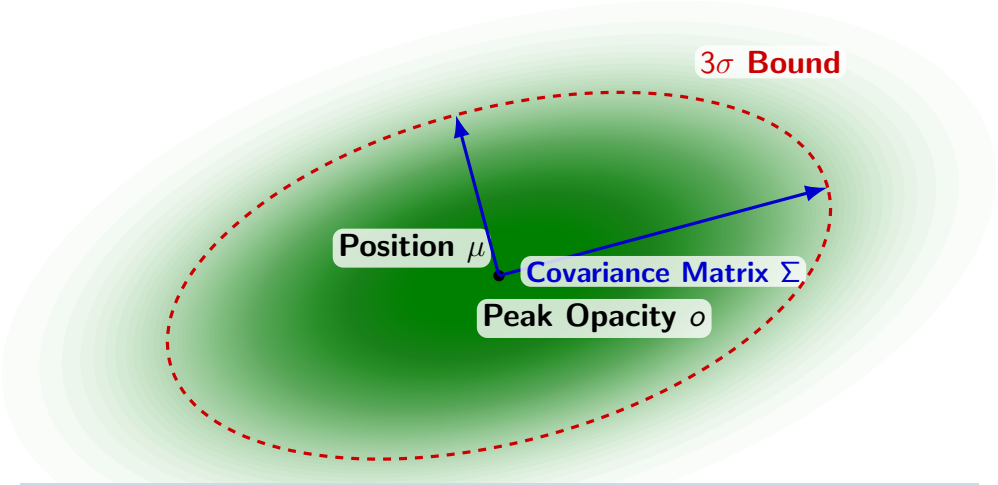

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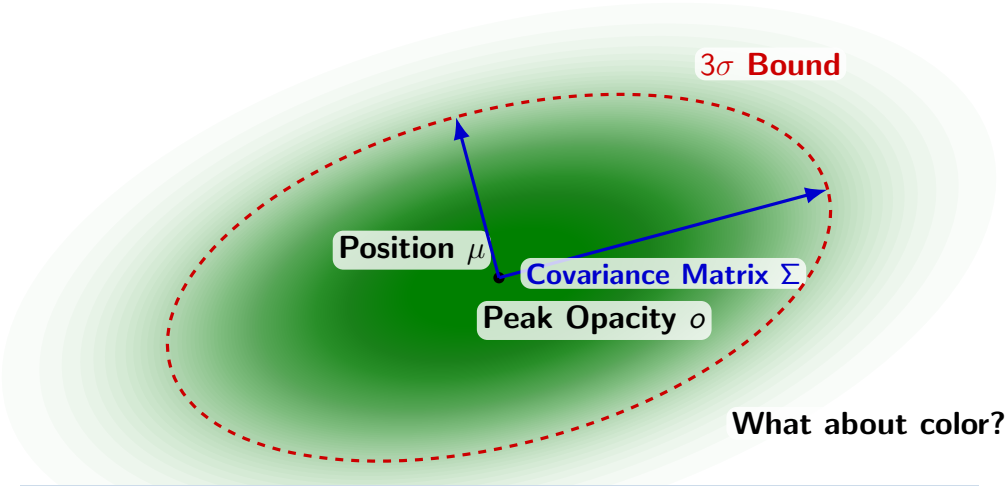

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Mathematical Setup

The Validity Problem

Core Decomposition

Scaling Values

Rotational Quaternion

Guaranteed Validity

Details

To work with gradient descent, the covariance matrix Σ must be differentiable and remain positive semi-definite (eigenvalues ≥ 0) to be geometrically valid.

Running a loop during backpropagation to find valid values is computationally too costly.

Mathematical Setup

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Details

Kerbl et al. bypassed this by decomposing Σ into a scaling matrix S and rotational matrix R mirroring standard eigendecomposition ($\Sigma = Q\Delta Q^T$):

$$\Sigma = RSS^TR^T \text{ with } \Delta = S^2, Q = R$$

Mathematical Setup

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Details

The diagonal scaling matrix S is further decomposed into three distinct scaling values (s_1, s_2, s_3). These represent the roots of the eigenvalues of Σ.

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Details

The rotational matrix R is decomposed into a four-value quaternion (w, x, y, z):

$$R = \begin{bmatrix} 1 - 2(y^2 + z^2) & 2(xy - wz) & 2(xz + wy) \\ 2(xy + wz) & 1 - 2(x^2 + z^2) & 2(yz - wx) \\ 2(xz - wy) & 2(yz + wx) & 1 - 2(x^2 + y^2) \end{bmatrix}$$

Mathematical Setup

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Details

3DGS only stores and optimizes these 7 specific values for each 3D Gaussian.

Because an eigendecomposition is always positive semi-definite, constructing a new covariance matrix from these components is guaranteed to be geometrically valid.

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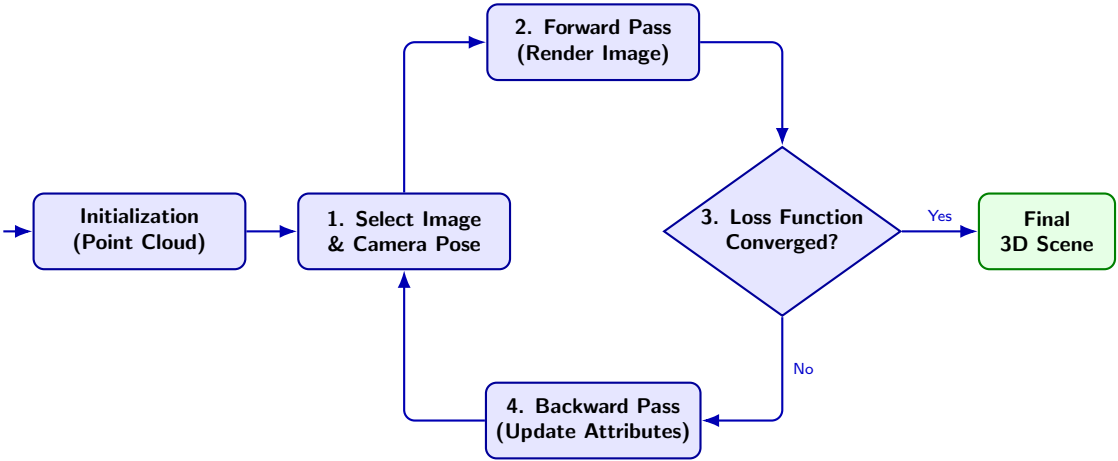
The Training Cycle



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Initialization Steps

Structure from Motion

Camera Matrix

Primitive Initialization

Details

The 3DGS algorithm requires initial geometry, which is commonly derived from a set of input images.

Because camera poses are usually unknown, Structure from Motion (SfM) software like COLMAP is used to solve a geometric optimization problem, providing an accurate 3D point cloud.

Initialization Steps

Structure from Motion

Camera Matrix

Primitive Initialization

Details

As another result from SfM, each input image is assigned a 3×3 camera matrix W_{rot} which contains the camera's orientation in world space.

Initialization Steps

Structure from Motion

Camera Matrix

Primitive Initialization

Details

The SfM points act as the origins for the initial 3D Gaussians.

- They are initialized as perfect spheres.
- Radii are deduced from their three nearest neighbors.
- Opacity is set to 0.1 and color to any RGB value.

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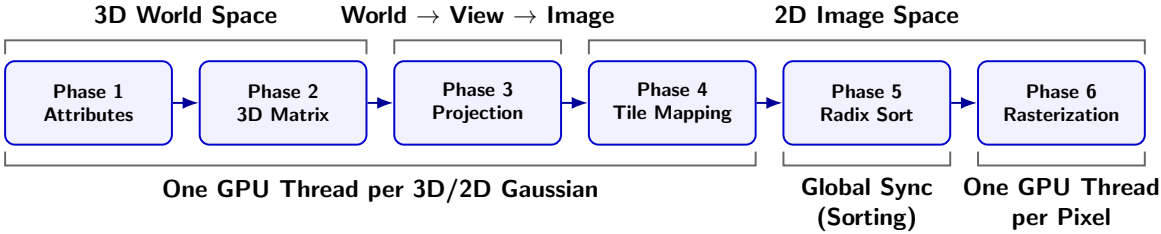
Forward Pass: Pipeline Architecture



Phase 1: Preparing Constituent Attributes

- The GPU is running one GPU thread per 3D Gaussian.
- Raw values optimized during training are converted from latent space back into physical space.
- Opacity is bounded to $(0, 1)$ using a sigmoid function, scale uses the exponential function, and the rotational quaternion is normalized.

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Phase 2: Reconstructing the 3D Covariance Matrix

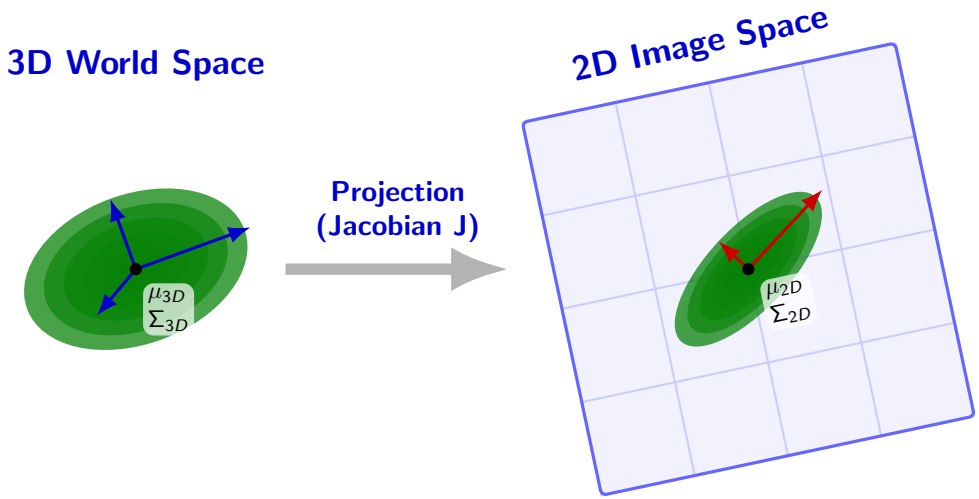
- The algorithm mathematically rebuilds the 3D geometric shape.
- It combines the scaling matrix S and rotational matrix R to form the full 3D covariance matrix:

$$\Sigma_{3D} = RSS^T R^T$$

Phase 3: Projection from 3D to 2D

- Converts 3D Gaussians (ellipsoids) in world space into flat 2D Gaussians (ellipses) in image space.
- Employs a linear affine approximation (the Jacobian matrix J) of a standard non-linear pinhole projection.
- The 3D matrix is rotated into view space via W_{rot} and then projected:

$$\Sigma_{2D} = J W_{rot} \Sigma_{3D} W_{rot}^T J^T$$


3D World Space
μ_{3D}
Σ_{3D}
Projection
(Jacobian J)
2D Image Space
μ_{2D}
Σ_{2D}

Phase 4: Tile Mapping (Preparing Radix Sort)

- The target output image is divided into a grid of 16×16 pixel tiles.
- The algorithm calculates a 2D bounding box to map exactly which tiles a given 2D Gaussian covers.
- Assigns each 2D Gaussian to a tile. If overlapping, assigns to all affected tiles.

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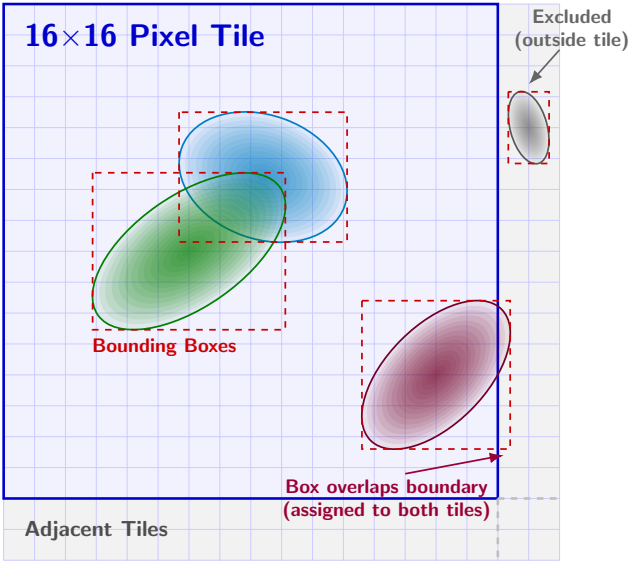
Phase 5: Global Radix Sort

- The GPU waits until all per-Gaussian computations are finished.
- A highly optimized Radix Sort is executed, organizing the 2D Gaussians for each tile strictly by depth (closest to farthest).

Forward Pass: Phases 4 & 5



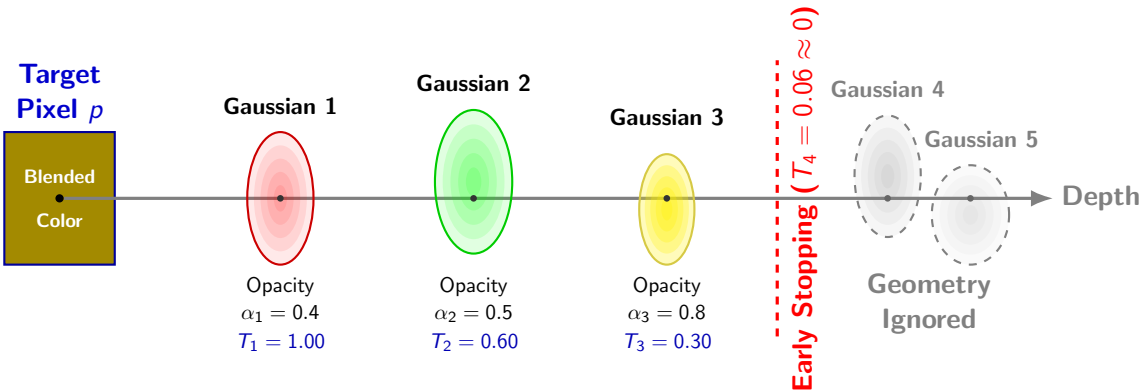
Phase 6: Tile-Based Rasterization (Splatting)

- Operating natively in 2D image space, the GPU assigns one processing thread per pixel.
- **Alpha Compositing:** Pixels accumulate color by blending the sorted list of overlapping 2D Gaussians front-to-back:

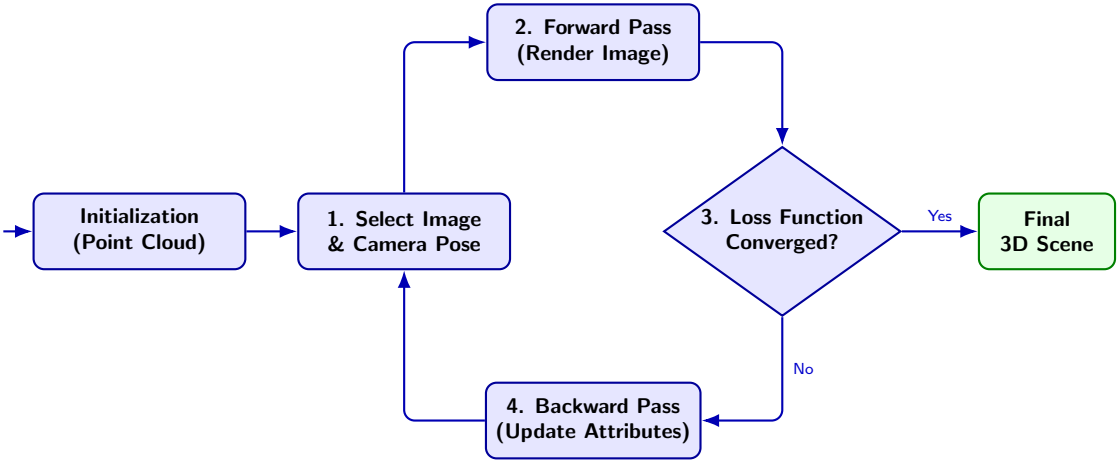
$$C_p = \sum_{i=1}^{N} c_i \alpha_i T_i, \quad \text{with} \quad T_i = \prod_{j=1}^{i-1} (1 - \alpha_j) \quad \text{the transmittance}$$

- **Early Stopping:** Threads stop calculating exactly when transmittance (T_i) drops near zero, ignoring hidden geometry for massive speed-ups.

Forward Pass: Phase 6



The Training Cycle



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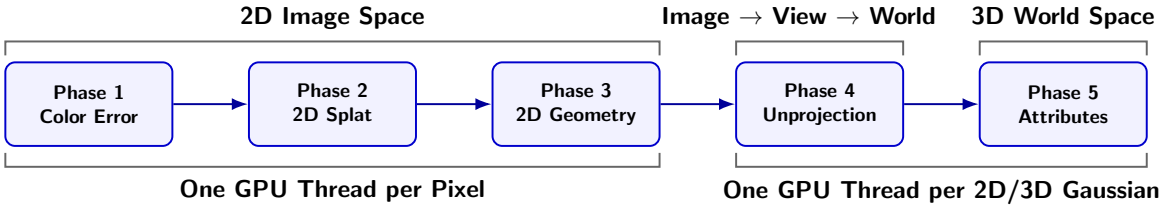
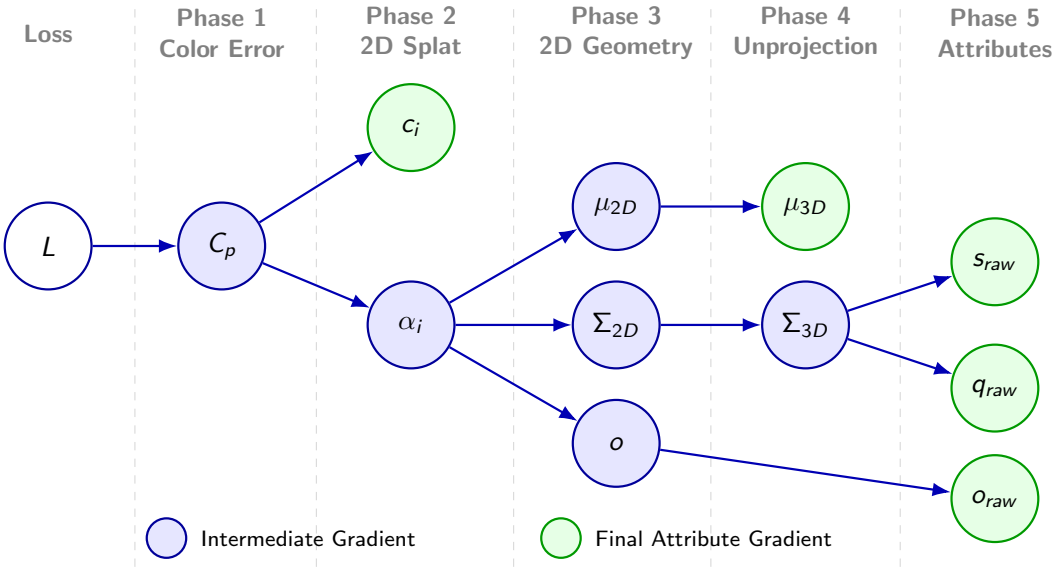
Backward Pass: Pipeline Architecture



Backward Pass: Gradient Flow Overview



The Loss Function

- Training compares the rendered image with the original input image using a weighted combination of $\mathcal{L}_1$ loss and D-SSIM.
- $\mathcal{L}_1$ calculates the absolute pixel-wise error, while D-SSIM acts as a structural regularizer:

$$L = (1 - \lambda) \cdot \mathcal{L}_1 + \lambda \cdot \mathcal{L}_{D-SSIM}, \quad \text{with } \lambda = 0.2$$

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Phase 1: Loss to Pixel Color Correction

- The gradient calculates the required color correction per pixel (C_p).
- A positive gradient indicates an overestimation of color, negative indicates underestimation, and the magnitude dictates the degree of adjustment required:

$$\frac{\partial L}{\partial C_p} = \frac{1 - \lambda}{N} \cdot sign(C_p - O_p) + \lambda \frac{\partial \mathcal{L}_{D-SSIM}}{\partial C_p}$$

Phase 2: Pixel Color Correction to 2D Splat

- The algorithm works alpha compositing backwards, flowing the pixel gradient into the evaluated opacity (α_i) and color (c_i) of each 2D Gaussian in the tile.

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- **Color Gradient:** Uses the chain rule to update the Gaussian's color based on its contribution (transmittance T_i and opacity α_i):

$$\text{with } C_p = \sum_{i=1}^{N} c_i \alpha_i T_i : \quad \frac{\partial L}{\partial c_i} = \frac{\partial L}{\partial C_p} \cdot \frac{\partial C_p}{\partial c_i} = \frac{\partial L}{\partial C_p} \cdot (\alpha_i T_i)$$

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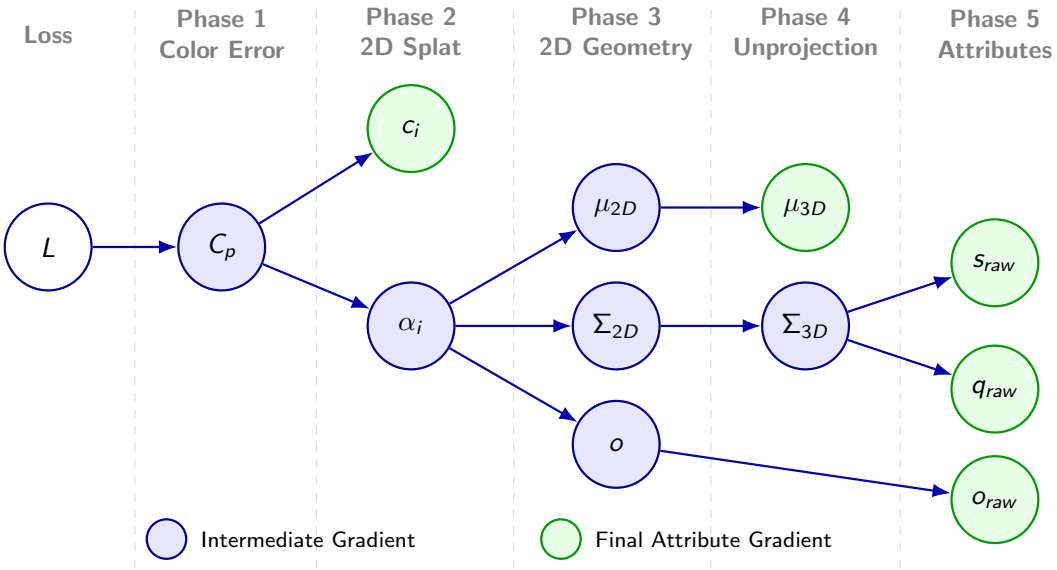
- **Opacity Gradient:** Increases evaluated opacity if the Gaussian's color (c_i) reduces the pixel error more effectively than the accumulated background color behind it (C_{behind}):

$$\frac{\partial L}{\partial \alpha_i} = \frac{\partial L}{\partial C_p} \cdot \frac{\partial C_p}{\partial \alpha_i} = \frac{\partial L}{\partial C_p} \cdot T_i(c_i - C_{behind})$$

Backward Pass: Gradient Flow Overview



Phase 3: 2D Splat to 2D Geometry

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- This mathematically pulls the center position of the 2D Gaussian towards the pixel and stretches or contracts its 2D shape to minimize the error.

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Phase 4: 2D Geometry to 3D Geometry

- The GPU model changes from pixel-based to running per-Gaussian calculations.
- The 2D gradients are unprojected back into 3D space, translating the 2D adjustments into 3D center movements and 3D volume changes.

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Phase 5: 3D Geometry to 3D Gaussian Attributes

- The gradients flow backward through the 3D covariance matrix assembly.
- Adjustments are mapped directly into the raw, latent space variables for scale, rotation, and base opacity.

Adaptive Density Control

- Steps in every 100 iterations to counter over-reconstruction (covering too much geometry) and under-reconstruction (too few Gaussians).
- **Splitting:** Over-reconstructing Gaussians are split into two smaller ones.
- **Cloning:** Under-reconstructing Gaussians are cloned.

Adaptive Density Control

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Pruning & Opacity Reset

- 3D Gaussians with extremely low opacity or unbound scale are routinely deleted.
- Every 3,000 iterations, the base opacity is reset to almost 0 for all Gaussians. This forces unimportant Gaussians to fade out completely and be pruned.

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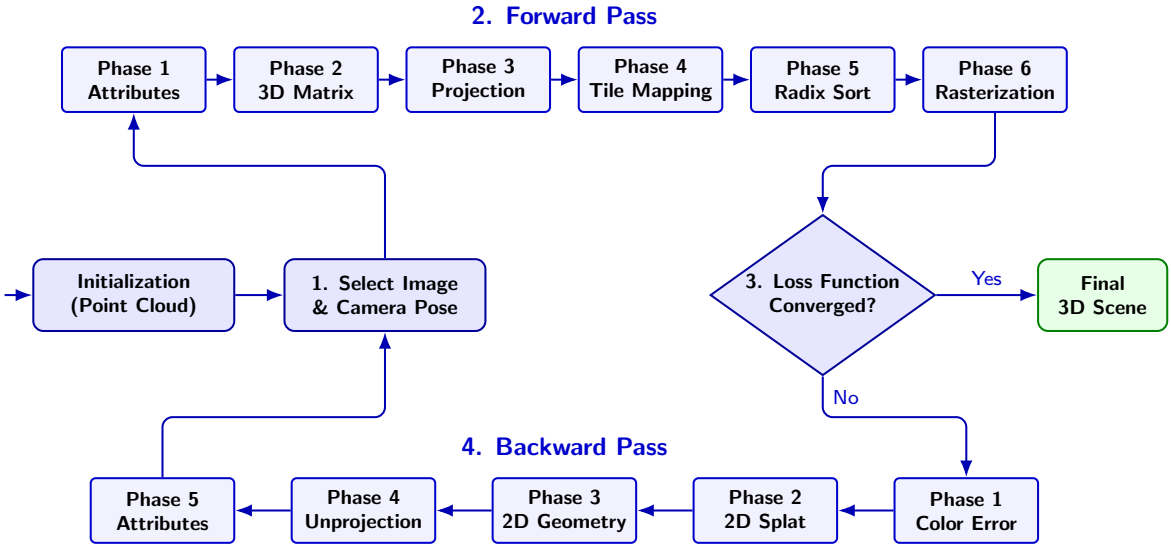
Optimization Step

- Rather than standard gradient descent, 3DGS utilizes the Adam optimizer.
- Adam takes each parameter's history into account, smoothing out updates so that 3D Gaussians gently morph into their correct geometric shapes.

The Detailed Training Cycle



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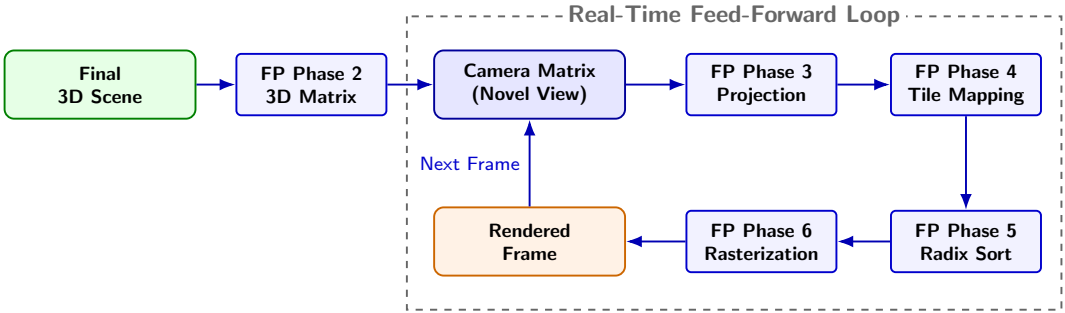
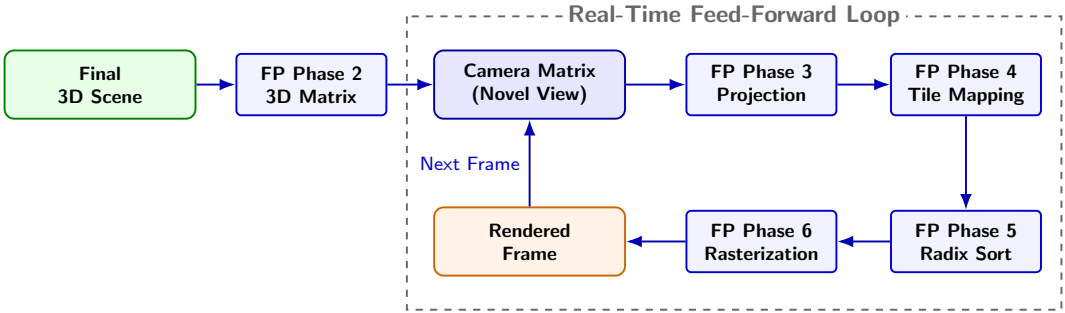
Novel-View Synthesis





- Because the scene geometry is finalized, the backward pass, Adaptive Density Control, and the Adam optimizer are completely disabled.
- Freed from the training overhead and powered by the raw speed of the tile-based rasterizer, 3DGS quickly reaches real-time rendering levels.

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Summary and Discussion

- 3D Gaussian Splatting is a learnable and explicit radiance field representation for 3D Reconstruction, abandoning neural networks.
- By combining multiple existing approaches and technologies, adding new ideas, and mapping them onto modern GPU hardware, Kerbl. et al made it possible to have real-time rendering of reconstructed 3D scenes for the first time.

I am looking forward to answering your questions.

Overview

6 Videos

7 Backpropagation Calculations

8 Selected References

Training in Progress

https://youtu.be/F9rgbvhSad8?t=550

https://youtu.be/yd40zUnr7eQ

Real-Time Rendering

Example from the original paper's project website

https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/content/videos/bicycle.mp4

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Phase 3: 2D Splat to 2D Geometry (Opacity & Position)

- We recall the simplified opacity distribution: $\alpha_i(p) = o \cdot e^E$, where E is the exponent derived from the pixel distance Δ and the inverse 2D covariance matrix.
- **Base Opacity (o):** The derivative scales directly with the distance from the center of the 2D Gaussian (e^E):

$$\frac{\partial L}{\partial o} = \frac{\partial L}{\partial \alpha_i} \cdot e^E$$

- **Position (μ_{2D}):** Let vector $v = \Sigma_{2D}^{-1}\Delta p$. This vector points directly from the 2D Gaussian's center to the pixel:

$$\frac{\partial L}{\partial \mu_{2D}} = \frac{\partial L}{\partial \alpha_i} \cdot \alpha_i(p) \cdot v$$

- If the evaluated opacity gradient is positive, the center is effectively pulled towards the pixel.

Phase 3: 2D Splat to 2D Geometry (Shape)

- **2D Covariance Matrix (Σ_{2D}):** We must calculate how the opacity at pixel p changes when the 2D shape of the Gaussian changes.
- This uses the same vector v from the position derivative, multiplied by its own transpose:

$$\frac{\partial L}{\partial \Sigma_{2D}} = \frac{\partial L}{\partial \alpha_i} \cdot \frac{1}{2} \alpha_i(p) \cdot (v \cdot v^T)$$

- Multiplying the vector by its transpose results in a matrix gradient.
- If the evaluated opacity gradient is positive, this matrix gradient causes the 2D Gaussian to stretch itself towards the pixel; otherwise, it causes it to contract.

Phase 4: 2D Geometry to 3D Geometry

- The GPU thread model changes to running per-Gaussian to unproject the 2D data back into 3D.
- This utilizes the multi-variable chain rule and the transposes of the forward projection matrices ($U = JW_{rot}$).
- **3D Position (μ_{3D}):** The gradient passes through view space (t) back to world space:

$$\frac{\partial L}{\partial \mu_{3D}} = W_{rot}^T J^T \frac{\partial L}{\partial \mu_{2D}} = U^T \left(\frac{\partial L}{\partial \mu_{2D}} \right)$$

- **3D Covariance (Σ_{3D}):** The final gradient unprojects cleanly using matrix U on both sides:

$$\frac{\partial L}{\partial \Sigma_{3D}} = U^T \left(\frac{\partial L}{\partial \Sigma_{2D}} \right) U$$

Phase 5: 3D Geometry to Matrix Components

- Gradients flow through the covariance matrix assembly ($\Sigma_{3D} = MM^T$ with $M = RS$) into scale (S) and rotation (R).
- First, the algorithm derives the intermediate matrix M:

$$\frac{\partial L}{\partial M} = 2 \left(\frac{\partial L}{\partial \Sigma_{3D}} \right) M$$

- Then, it splits this gradient into the separate components:

$$\frac{\partial L}{\partial S} = R^T \left(\frac{\partial L}{\partial M} \right) \quad \text{and} \quad \frac{\partial L}{\partial R} = \left(\frac{\partial L}{\partial M} \right) S$$

- The scale gradient ($\frac{\partial L}{\partial s}$) is extracted from the diagonal of $\frac{\partial L}{\partial S}$, while the quaternion gradient ($\frac{\partial L}{\partial q}$) aggregates the partial derivatives of R.

Phase 5: Backpropagating to Latent Attributes

- To close the loop, the gradients pass backwards through their respective activation functions to reach the raw, latent space variables.
- **Scale (Exponential):** Backpropagating through exp requires multiplying by s:

$$\frac{\partial L}{\partial s_{raw}} = \frac{\partial L}{\partial s} \cdot s$$

- **Rotation (Normalization):** Uses the Jacobian matrix ($J_{||}$) of the normalization function, with $J_{||} = \frac{I - qq^T}{||q_{raw}||}$:

$$\frac{\partial L}{\partial q_{raw}} = J_{||} \cdot \frac{\partial L}{\partial q}$$

- **Base Opacity (Sigmoid):** Backpropagating through the sigmoid function:

$$\frac{\partial L}{\partial o_{raw}} = \frac{\partial L}{\partial o} \cdot o(1 - o)$$

Overview

6 Videos

7 Backpropagation Calculations

8 Selected References

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